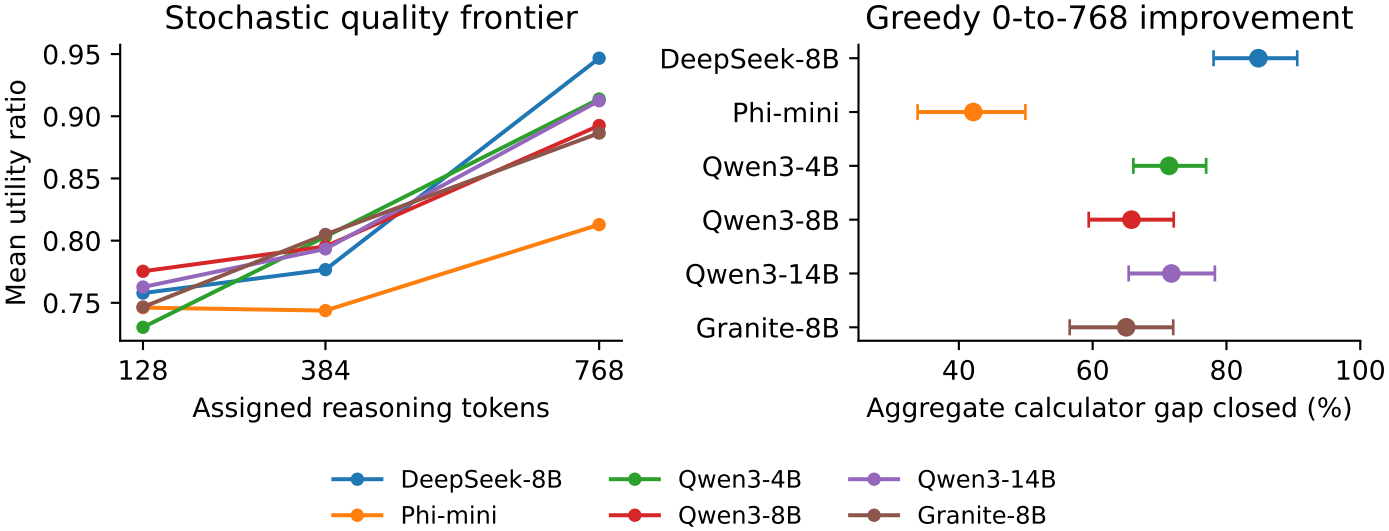
Stochastic quality frontier
Mean utility ratio
0.95
0.90
0.85
0.80
0.75
128
384
768
Assigned reasoning tokens
Greedy 0-to-768 improvement
DeepSeek-8B
Phi-mini
Qwen3-4B
Qwen3-8B
Qwen3-14B
Granite-8B
40
60
80
100
Aggregate calculator gap closed (%)
DeepSeek-8B
Phi-mini
Qwen3-4B
Qwen3-8B
Qwen3-14B
Granite-8B